

BANARAS HINDU UNIVERSITYB.Sc. (Hons) — Semester II Examination, 2013-14 (*Old Course*)**Paper No. MTB-202: Multivariable Calculus**

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Give the ε - δ definition of the limit of a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ as $(x, y) \rightarrow (a, b)$. Using the definition, show that:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}$$

does not exist.

[3]

- (b) Show that the simultaneous limit of $f(x, y) = \frac{xy}{x^2 + y^2}$ does not exist at $(0, 0)$, but that the iterated limits exist and are equal.

[2]

- (c) Let $f(x, y)$ be a homogeneous function of degree 5. Show that:

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 20f$$

[2]

- (d) State Taylor's theorem for functions of two variables.

[2]

- (e) Evaluate $\int_0^\infty x^n e^{-x} dx$.

[2]

- (f) Show that $\int_0^\infty e^{-x^2} dy = \Gamma\left(\frac{n+1}{2}\right)$, $n > 0$.

[2]

- (g) Write a necessary condition for $f(x, y)$ to have an extremal value.

[2]

- (h) If x, y, z are positive and $h_1 \leq x + y + z \leq h_2$, then find:

$$\iiint (x + y + z) x^m y^n z^l dx dy dz$$

[2]

- (i) If $\vec{r}$ is a vector function of scalar t and $\vec{a}$ is a constant vector, differentiate the following with respect to t :

(i) $\vec{r} \cdot \vec{a}$

(ii) $\vec{r} \times \vec{a}$

(iii) $|\vec{r}| \frac{d\vec{r}}{dt}$

[3]

(j) Prove that $\operatorname{div} \vec{r} = 3$.

[2]

Question 2.

(a) Prove that the limit of the product of two functions of two variables equals the product of their limits. [7]

(b) Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at the origin.

[5]

Question 3.(a) If (a, b) is a point of the domain $D \subset \mathbb{R}^2$ of a real-valued function $f(x, y)$, such that (i) $f_x(a, b)$ exists, (ii) $f_y(x, y)$ is continuous at (a, b) , then prove that $f(x, y)$ is differentiable at (a, b) . [7](b) Show that $f(x, y) = \sin x + \cos y$ is differentiable everywhere.

[5]

Question 4.(a) If A, B, C are the angles of a triangle ABC , find the maxima and minima of $u = \sin A \cdot \sin B \cdot \sin C$. [7](b) Transform the integral $\int_0^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{x^2 + y^2} dx dy$ by changing to polar coordinates and hence evaluate it. [5]**Question 5.**(a) Show that $B(m, n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy$. [6](b) By changing the order of integration, evaluate $\int_0^{\pi/2} \int_0^{2a \cos \theta} f(r, \theta) dr d\theta$. [6]**Question 6.**(a) (i) If $\vec{a}$ is a constant vector, find $\operatorname{div}(\vec{r} \times \vec{a})$ and $\operatorname{curl}(\vec{r} \times \vec{a})$.

(ii) Prove that $\nabla\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$.

[6]

(b) Prove that $\nabla(\vec{a} \cdot \vec{b}) = \vec{a} \times \text{curl } \vec{b} + \vec{b} \times \text{curl } \vec{a} + (\vec{a} \cdot \nabla)\vec{b} + (\vec{b} \cdot \nabla)\vec{a}$.

[6]

Question 7.

(a) Verify Stokes' theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken round the rectangle bounded by $x = \pm a, y = 0, y = b$.

[6]

(b) State and prove the Gauss Divergence Theorem.

[6]